

Course Code : ECT 453-2

KQLR/RS – 24 / 1679

**Seventh Semester B. Tech. (Electronics and Communication
Engineering) Examination**

LONG TERM EVOLUTION TECHNOLOGIES

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) Solve all questions.
- (2) Draw diagrams wherever necessary.

1.
 - (a) Describe the functionalities of LTE Network Architecture along with the block diagram. 7(CO1)
 - (b) What is MIMO ? 3(CO2,3)
2.
 - (a) Demonstrate the elegance of multicarrier modulation in OFDM for Delay Spread, Inter symbol Interference and Frequency Selective fading. 7(CO3,4)
 - (b) What is the minimum and maximum data rate loss due to the cyclic prefix ? What is the minimum theoretical loss if a shorter cyclic prefix could be used ? 3(CO3,4)
3.
 - (a) Why two different modulation techniques are used in LTE uplink and LTE downlink ? Explain in brief. 5(CO2,3)
 - (b) How many bits are delivered by a Physical Resource Block (PRB) with normal Cyclic Prefix (CP) where all subcarriers are modulated with 16 QAM ? Why ? 5(CO2,3)

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Contd.

4. (a) Elucidate the LTE Resource block structure, with its time duration specifications. 5(CO4)
- (b) Illustrate the radio interface protocol architecture and the SAPs between different layers of LTE. 5(CO1,3)
5. (a) Give an account of PDCP overview. 5(CO4)
- (b) Illustrate the Cell Search as deployed in LTE. 5(CO3,4)
6. How LTE-A technology is different than LTE and its predecessors ? 10(CO5)

